Portland Arithmetic Stress Task

Instruction Manual - intensified build, 50% target error rate

1. What this program is

This is a Python implementation of the **Portland Arithmetic Stress Task (PAST)**, a laboratory stressor built for event-related potential research. Participants solve mental arithmetic under time pressure while the difficulty adjusts continuously to hold them near a fixed failure rate, and while a simulated peer average is displayed beside their score.

Original paradigm: Atchley, Ellingson, Klee, Memmott & Oken (2017). *A cognitive stressor for event-related potential studies: the Portland arithmetic stress task.* Stress, 20(3), 277-284. The original was written in E-Prime; this is an independent Python reimplementation.

> **This build is deliberately harder than the published task.** The published PAST titrates difficulty to hold the error rate near **40 percent**. This build targets **50 percent**, by advancing the difficulty as soon as three of five trials are correct instead of more than three of five. Every other parameter follows the published description. Because the failure rate differs, results from this build are not directly comparable to published PAST norms without noting the change. The deviation is recorded in every session.json file and is stated to the participant on screen before the task begins.

2. Installing

You need **Python 3.10 or newer**. During the Python installation, tick *Add python.exe to PATH*.

1. Double-click **install.bat** in this folder.
2. Wait for it to finish. It downloads a few hundred megabytes, so allow several minutes.
3. It ends by running a self-test. You should see **RESULT: PASS**.

The installer builds a private environment in a **.venv** folder. This is deliberate: PsychoPy has a large dependency tree and some releases require older versions of numpy, which could disturb other Python tools on the same computer. Keeping it separate avoids that entirely.

To install manually instead, open a terminal in this folder and run:

```
pip install -r requirements.txt
```

3. Running a session

1. Double-click **run.bat**. The setup form appears on the task screen.
2. Use **UP / DOWN** to choose a field, **type** to edit the participant ID and session, **LEFT / RIGHT** to change the language, the mode and the minutes, then press **ENTER** to start.
3. The task takes over the whole screen. Hand the keyboard to the participant.
4. **ESC stops the session at any moment** and saves everything collected so far.

Setting	Meaning
Participant ID	Used in the output folder name. Anything without spaces works.
Participant language	English or ■■. Fixed for the whole session so wording never changes mid-task.
Session mode	Standard or Timed - see section 4.
Timed mode: total minutes	Only used in Timed mode (default 8). Ignored in Standard mode.
Full screen	On by default. Pass --windowed on the command line for testing.

4. The two session modes

Mode	What it does	Use it when
Standard PAST	2 runs of 80 trials, exactly as published, with a rest between runs for a blood-pressure measurement. Roughly 12 to 16 minutes of task.	You want trial counts comparable to the published paradigm.
Timed	You enter a total number of minutes. The task runs on the clock and stops when the time is up, so the trial count varies with how fast the participant works.	You need every participant to receive the same duration of stress, or you must fit a fixed recording window.

5. What the participant does

A sum appears in the centre of the screen. It must be worked out **strictly left to right**; the usual precedence rule that multiplication comes before addition does **not** apply. For example **54 - 53 x 7** means (54 - 53) x 7, which is 7. This is explained on screen and in the practice trials.

Answers are entered on a number wheel with three keys, reproducing the three-button response box used in the original study:

Key	Action
RIGHT arrow	Turn the wheel clockwise
LEFT arrow	Turn the wheel counterclockwise
ENTER or SPACE	Select the highlighted item
OK	Submit the answer
DEL	Delete the last digit
+/-	Toggle a negative answer
ESC	Stop the session immediately

The green bar at the top fills as time runs out, turning amber then red. If it completes before an answer is submitted, the trial is scored as an error.

6. How the difficulty adapts

Performance is reviewed after every five trials and the task moves up or down a five-rung ladder. Each rung changes both the arithmetic complexity and the time allowed.

Correct in the last 5	What happens
3, 4 or 5	Harder: one level up, shorter time window
0, 1 or 2	Easier: one level down, longer time window

Level	Problem shape	Thinking time
0	2 numbers, 1 digit, + and -	9.0 s
1	2 numbers, 2 digits, + and -	7.5 s
2	3 numbers, 2 digits, + - x	6.0 s
3	3 numbers, 2 digits, + - x /	5.0 s
4	4 numbers, 2 digits, + - x /	4.0 s

> The times above are the budget for **working the sum out**. On top of that, every trial is given an allowance for keying the answer into the wheel, calculated from the minimum number of presses that particular answer needs. Without it a long answer such as -487 would fail for motor reasons rather than arithmetic ones, and the difficulty would end up chasing an artefact of the input device instead of the participant's arithmetic. Both components are logged per trial as think_s and entry_s.

> Multipliers and divisors are always single digits, and every answer is a whole number between -999 and 999. Two-digit multiplication is beyond mental arithmetic for nearly everyone; an impossible item would measure nothing and only add noise, so the generator does not produce them.

7. EEG synchronisation

The task publishes an **LSL marker stream named PAST_Markers**. Start your EEG recording and have it subscribe to that stream, or record it with LabRecorder alongside the EEG. Markers are also written to **markers.csv** with the same timestamps, so a session is still analysable if LSL was not running.

For feedback-related negativity, epoch around the feedback markers - the published analysis window is 200 ms before to 1000 ms after feedback onset.

Code	Marker	Sent when
200	SESSION_START	Session begins
203 / 204	BASELINE_START / END	Quiet baseline period
205 / 206	RUN_START / END	Each run
210	TRIAL_ONSET	Problem appears
220	RESPONSE	Answer submitted
230	FEEDBACK_CORRECT	Correct feedback onset
231	FEEDBACK_INCORRECT	Incorrect feedback onset
232	FEEDBACK_TIMEOUT	Timeout feedback onset
240 / 241 / 242	TITRATION_UP / DOWN / HOLD	Difficulty adjusted
250 / 251	REST_START / END	Rest between runs
201 / 202, 260 / 261	PANAS_PRE, PANAS_POST	Questionnaire start and end
265	VAS_RATING	Stress slider shown
298 / 299	ABORTED / SESSION_END	How the session ended

8. Questionnaires

Two measures are collected before and after the task, as a manipulation check that stress was actually induced:

Stress VAS - a 0 to 100 slider, also shown between runs.

State PANAS - the 20-item Positive and Negative Affect Schedule (Watson, Clark & Tellegen, 1988), rated 1 to 5 for how the participant feels right now. Positive and negative subscale totals (each 10 to 50) are computed automatically and written to session.json.

9. Output files

Every session writes a new timestamped folder under **data**, so nothing is ever overwritten.

File	Contents
trials.csv	One row per trial: the problem, the correct answer, the response, correctness, reaction time, difficulty level, the time limit and its thinking/entry split, running accuracy, the peer figure shown, and any titration event. Written as the session runs, so a crash cannot lose it.
markers.csv	Every marker with its task timestamp and wall-clock time.
session.json	How the session was configured, the summary results, the PANAS and VAS scores, the full titration history, and an explicit record of the 50 percent deviation from the published task.

10. Running this ethically

This task works by making people fail and by making them feel judged for it. That is the point of it, and it produces a real physiological stress response. Treat it as you would any stress-induction protocol.

1. The program shows the participant, before the task begins, that it is designed to be stressful, that about half their answers will be wrong by design, and that they may stop at any time with ESC.
2. The debrief screen at the end is **not optional**. It discloses that the failure rate was engineered and that the group average was simulated. Do not skip it, and do not end a session before a participant has read it.
3. If a participant stops early with ESC, still show them the debrief and explain what the task was doing.
4. Obtain ethics approval before running human participants, and follow whatever your institution requires for stress induction.

11. If something goes wrong

Symptom	What to do
The libraries are not installed yet	Run install.bat first. run.bat needs the .venv folder to exist.
LSL markers: unavailable	The task still runs and still writes markers.csv. Reinstall pylsl if you need live streaming.
No sound	Check session.json for audio_error. The task runs silently rather than stopping, but the auditory feedback is part of the stressor, so fix it before collecting real data.
The window is the wrong size	Untick Full screen in the setup dialog, or pass --windowed.
Everything is too hard or too easy	Edit config.py. The difficulty ladder and the titration thresholds are all near the top of that file.

12. Checking the task without a participant

This runs the arithmetic generator and the titration algorithm against a simulated participant and prints the resulting error rate. It needs no display and takes a second:

```
.venv\Scripts\python main.py --selftest
```

The error rate should land between 40 and 60 percent across simulated ability levels, centred near 50. If it does not, the titration settings in config.py have been changed.

波特兰算术压力任务

使用说明书 - 强化版本，目标错误率 50%

1. 本程序简介

本程序是 波特兰算术压力任务 (PAST) 的 Python 实现。该任务是一种为事件相关电位研究设计的实验室压力源。被试需在时间压力下完成心算，任务难度会持续调整以使其保持在固定的失败率附近，同时在其成绩旁显示一个模拟的同伴平均分。

原始范式来源：Atchley, Ellingson, Klee, Memmott 与 Oken (2017)。A cognitive stressor for event-related potential studies: the Portland arithmetic stress task. Stress, 20(3), 277-284。原版使用 E-Prime 编写，本程序为独立的 Python 重新实现。

> 本版本刻意比已发表版本更难。已发表的 PAST 将错误率调控在 40% 左右；本版本目标为 50%，做法是把「五题中答对超过三题才升难度」改为「五题中答对三题即升难度」。其余参数均遵循已发表的描述。由于失败率不同，本版本的结果若要与已发表的 PAST 常模比较，必须注明这一改动。该偏差会记录在每个 session.json 文件中，并在任务开始前于屏幕上告知被试。

2. 安装

需要 Python 3.10 或更新版本。安装 Python 时请勾选 Add python.exe to PATH。

1. 双击本文件夹中的 install.bat。
2. 等待安装完成。需下载数百 MB，可能耗时数分钟。
3. 安装结束时会自动运行自检，应显示 RESULT: PASS。

安装程序会在 .venv 文件夹中建立一个独立环境。这样做是有意为之：PsychoPy 依赖库庞大，某些版本需要较旧的 numpy，可能影响同一台电脑上的其他 Python 工具。使用独立环境可完全避免此问题。

若要手动安装，请在本文件夹打开终端并执行：

```
pip install -r requirements.txt
```

3. 运行实验

1. 双击 run.bat，设置表单会显示在任务屏幕上。
2. 用 上 / 下 方向键选择字段，直接输入 编辑被试编号与场次，左 / 右 方向键切换语言、模式与分钟数，然后按 回车 开始。
3. 任务将全屏显示，此时把键盘交给被试。
4. 任何时候按 ESC 都可停止实验，并保存已采集的全部数据。

设置项	说明
被试编号	用于输出文件夹命名，建议不含空格。
被试语言	English 或 中文。整场实验固定不变，以免任务中途措辞改变。
实验模式	标准模式或计时模式，详见第 4 节。
计时模式：总分钟数	仅在计时模式下生效（默认 8 分钟），标准模式下忽略。
全屏	默认开启。测试时可在命令行加上 --windowed。

4. 两种实验模式

模式	说明	适用情形
标准 PAST	2 轮，每轮 80 试次，与已发表方案一致，两轮之间设休息以便测量血压。任务时长约 12 至 16 分钟。	需要与已发表范式可比的试次数量时。
计时模式	由你输入总分钟数。任务按时钟运行，时间到即结束，因此试次数量随被试作答速度而变化。	需要每位被试承受相同时长的压力，或需配合固定的记录时间窗时。

5. 被试的操作

屏幕中央会出现一道算式，必须 严格从左到右 计算，不 使用「先乘除后加减」的通常规则。例如 54 − 53 × 7 表示 (54 − 53) × 7，结果为 7。屏幕说明与练习试次中都会说明这一点。

使用三个按键在数字转盘上输入答案，以复现原研究所用的三键作答盒：

按键	作用
右方向键	顺时针转动转盘
左方向键	逆时针转动转盘
回车 或 空格键	选中当前高亮项
OK	提交答案
DEL	删除最后一位数字
+/-	切换答案正负号
ESC	立即停止实验

顶部的绿色进度条随时间流逝而填充，依次变为琥珀色与红色。若在提交答案前进度条填满，该试次记为错误。

6. 难度如何自适应

每五个试次评估一次成绩，任务在五级难度阶梯上上调或下调。每一级都会同时改变算术复杂度与允许时间。

最近 5 题答对数	结果
3、4 或 5	变难：升一级，时间窗缩短
0、1 或 2	变简单：降一级，时间窗延长

级别	题目结构	思考时间
0	2 个数，1 位数，+ 与 −	9.0 s
1	2 个数，2 位数，+ 与 −	7.5 s
2	3 个数，2 位数，+ − ×	6.0 s
3	3 个数，2 位数，+ − × ÷	5.0 s
4	4 个数，2 位数，+ − × ÷	4.0 s

> 上表的时间是 计算题目 的预算。在此之外，每个试次还会额外获得在转盘上输入答案的时间补偿，其数值由该答案所需的最少按键次数算出。若无此补偿，像 −487 这样的长答案会因手部操作而非算术能力而失败，难度调整也会追逐输入设备的假象而非被试的真实算术水平。两部分时间都会以 think_s 与 entry_s 逐试次记录。

> 乘数与除数始终为一位数，所有答案均为 −999 至 999 之间的整数。两位数相乘对绝大多数人而言已超出心算能力，无法完成的题目不能测量任何东西，只会增加噪声，因此题目生成器不会产生此类题目。

7. 脑电同步

任务会发布一个名为 PAST_Markers 的 LSL 标记流。请启动脑电记录并订阅该流，或使用 LabRecorder 与脑电一同记录。标记同时以相同时间戳写入 markers.csv，因此即使 LSL 未运行，该场次仍可分析。

分析反馈相关负波时，请围绕反馈标记分段；已发表的分析窗口为反馈起始前 200 毫秒至其后 1000 毫秒。

代码	标记	发送时机
200	SESSION_START	实验开始
203 / 204	BASELINE_START / END	静息基线时段
205 / 206	RUN_START / END	每一轮开始与结束
210	TRIAL_ONSET	题目出现
220	RESPONSE	提交答案
230	FEEDBACK_CORRECT	正确反馈出现
231	FEEDBACK_INCORRECT	错误反馈出现
232	FEEDBACK_TIMEOUT	超时反馈出现
240 / 241 / 242	TITRATION_UP / DOWN / HOLD	难度调整
250 / 251	REST_START / END	轮间休息
201 / 202, 260 / 261	PANAS_PRE, PANAS_POST	问卷开始与结束
265	VAS_RATING	压力滑条出现
298 / 299	ABORTED / SESSION_END	实验结束方式

8. 问卷

任务前后各采集两项测量，作为压力确实被诱发的操作检验：

压力 VAS - 0 至 100 的滑条，轮间也会呈现。

状态 PANAS - 20 项正负情绪量表（Watson, Clark 与 Tellegen, 1988），按 1 至 5
评定此刻的感受。正性与负性分量表总分（各为 10 至 50）会自动计算并写入 session.json。

9. 输出文件

每场实验都会在 data 下建立一个带时间戳的新文件夹，不会覆盖任何既有数据。

文件	内容
trials.csv	每试次一行：题目、正确答案、作答、正误、反应时、难度级别、时间限制及其思考/输入拆分、累计正确率、显示的同伴分数，以及任何难度调整事件。边运行边写入，程序崩溃也不会丢失。
markers.csv	全部标记及其任务时间戳与真实时钟时间。
session.json	实验的配置、结果摘要、PANAS 与 VAS 得分、完整的难度调整历史，以及关于「与已发表任务相差 50%」的明确记录。

10. 伦理注意事项

本任务的作用机制是让被试不断失败，并让他们因此感到被评价。这正是其设计目的，并会引发真实的生理应激反应。请像对待任何压力诱发方案一样谨慎使用。

1. 程序会在任务开始前告知被试：该任务被设计为具有压力性，约有一半答案按设计会是错误的，且可随时按 ESC 停止。
2. 结束时的实验说明屏幕 不可省略。它会告知失败率是人为设定的，且小组平均分是模拟的。请勿跳过，也不要在被试阅读之前结束实验。
3. 若被试提前按 ESC 停止，仍应向其展示实验说明并解释任务的机制。
4. 在招募人类被试前须取得伦理审批，并遵守所在机构对压力诱发研究的各项要求。

11. 常见问题处理

现象	处理方法
提示尚未安装库	请先运行 install.bat。run.bat 需要 .venv 文件夹存在。
LSL 标记不可用	任务仍可运行并写入 markers.csv。若需实时streaming，请重新安装 pylsl。
没有声音	请检查 session.json 中的 audio_error。任务会静音继续而不会中止，但听觉反馈是压力源的一部分，正式采集数据前请先修复。
窗口尺寸不对	在设置对话框中取消勾选全屏，或加上 --windowed 参数。
整体过难或过易	编辑 config.py。难度阶梯与调整阈值都位于该文件靠前的位置。

12. 无被试情况下的检查

以下命令会用模拟被试运行题目生成器与难度调整算法，并打印所得的错误率。无需显示器，约一秒完成：

```
.venv\Scripts\python main.py --selftest
```

在各模拟能力水平下，错误率应落在 40% 至 60% 之间，并接近 50%。若不在此范围，说明 config.py 中的调整参数已被修改。